Below is a **very practical "day-to-day used" Microservices Classes + Methods list** (Java ecosystem focused: **Spring Boot + Spring Cloud + REST + DB + Kafka + Security + Observability**) with **real-time use cases + examples**, in the form of a **table**.

✅ This is NOT "all classes in Spring" (impossible).
✅ This is **what developers actually use daily** while building real microservices in production.

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1) REST API Layer (Controller + Web)

Class / Annotation	Most-used Methods	Real-time Use Case	Example
@RestController	(No methods, annotation)	Create REST endpoints returning JSON	@RestController class UserController {}
@RequestMapping	value, method	Base URL mapping for controller	@RequestMapping ("/users")
@GetMapping	mapping	Fetch data	GET /users/1
@PostMapping	mapping	Create data	POST /users
@PutMapping	mapping	Full update	PUT /users/1
@PatchMapping	mapping	Partial update	PATCH /users/1
@DeleteMapping	mapping	Delete resource	DELETE /users/1
@RequestParam	required, defaultValue	Filter/pagination query params	/users?page=0&size=10
@PathVariable	binding	Read URL variable	/users/{id}
@RequestBody	binding	Read JSON request body	createUser (@RequestBody UserDTO dto)
@ResponseStatus	status	Send correct HTTP status	@ResponseStatus (HttpStatus.CREATED)
ResponseEntity<T>	ok(), status(), body()	Customize status + headers	ResponseEntity.status (201).body(x)
HttpStatus	status codes	Standard REST behavior	400, 404, 500

HttpHeaders	add(), set()	Set JWT, correlation id, caching	headers.add ("X-Req-Id", id)
ServletRequest/Response	getters	low-level access	read remote IP, headers
@CrossOrigin	allowed origins	Frontend integration	React/Angular to service
MultipartFile	getBytes(), getInputStream(), getOriginalFilename()	Upload documents/images	KYC document upload

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2) Validation (Very common)

Class / Annotation	**Key Methods**	**Real-time Use Case**	**Example**
@Valid	triggers validation	Validate DTO inputs	Create (@Valid @RequestBody dto)
@NotNull	validation	mandatory field	@NotNull Long userId;
@NotBlank	validation	required string	@NotBlank String name;
@Size	validation	length constraints	@Size (min=8,max=20)
@Pattern	regex	PAN, email, phone validation	@Pattern (regexp="...")
@Min/@Max	numeric	limits	age range
BindingResult	hasErrors(), getAllErrors()	custom validation messages	show field errors
MethodArgumentNotValidException	exception	global exception handling	invalid request

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3) Exception Handling (Production MUST)

Class / Annotation	Methods	Real-time Use Case	Example
@ControllerAdvice	global handler	Central error handling	all controllers
@ExceptionHandler	mapping exceptions	handle custom exception	@ExceptionHandler (UserNotFoundException.class)
ResponseStatusException	constructors	Throw proper HTTP error	throw new ResponseStatusException(404,"Not Found")
ProblemDetail (Spring 6+)	setters	standard RFC7807 error response	proper error JSON
Custom Exception class	constructor	business errors	InsufficientBalanceException

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4) Service Layer (Business Logic)

Class / Annotation	Key Methods	Use Case	Example
@Service	annotation	mark business service	@Service class UserService {}
@Transactional	annotation	atomic DB operations	money transfer
@Async	async execution	email, notifications	async send mail
CompletableFuture	supplyAsync(), thenApply(), thenCompose(), allOf()	parallel microservice calls	calling multiple services
Optional<T>	ofNullable(), orElse(), orElseThrow(), isPresent()	avoid NPE	repo.findById(id).orElseThrow()

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5) Data Access Layer (JPA / Hibernate)

Class / Annotation	Key Methods	Real-time Use Case	Example
JpaRepository<T,ID>	save(), findById(), findAll(), deleteById(), existsById()	CRUD operations	user CRUD
CrudRepository	CRUD	minimal repository	simple service
PagingAndSortingRepository	findAll(Pageable)	pagination	admin list
@Entity	mapping	DB table mapping	UserEntity
@Table	table mapping	custom table	@Table (name="users")
@Id	primary key	PK config	@Id Long id;
@GeneratedValue	ID generation	auto ID	Strategy
@Column	constraints	column control	unique, length
@OneToMany/@ManyToOne	mapping	relations	Orders ↔ User
Page<T>	getContent(), getTotalPages()	paginated response	list endpoint
PageRequest	of(page,size,sort)	pagination	PageRequest.of (0,10)
Sort	by(), ascending()	sorting	sort by createdAt
@Query	custom JPQL	complex query	@Query ("select...")

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6) Database Transaction & JDBC (Used Often)

Class	Key Methods	Use Case	Example
JdbcTemplate	query(), queryForObject(), update(), batchUpdate()	High performance SQL	reporting service
NamedParameterJdbcTemplate	query(), update()	named params	flexible queries
RowMapper<T>	mapRow()	map result to object	custom mapping
DataSource	connection provider	DB connection pool	HikariCP

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7) Microservices Communication (RestTemplate / WebClient / Feign)

Class	Key Methods	Use Case	Example
RestTemplate (legacy)	getForObject(), postForEntity(), exchange()	sync calls	call inventory service
WebClient (modern)	get(), post(), retrieve(), bodyToMono()	reactive non-blocking calls	payments, streaming
Mono/Flux	map(), flatMap(), zip()	reactive chains	pipeline processing
@FeignClient	interface mapping	clean service-to-service calls	@FeignClient ("user-service")
FeignException	status()	handle downstream errors	fallback logic

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8) API Gateway / Routing (Spring Cloud Gateway)

Class/Config	Key Methods	Real-world Use Case	Example
Route config	predicates/filters	route to services	/user/** -> user-service

Gateway filters	modify request/response	auth, correlation id	add headers
Circuit breaking filters	resilience	stop cascading failures	breaker fallback

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9) Service Discovery + Config (Spring Cloud)

Component	Methods	Use Case	Example
Eureka Client	registers service	service discovery	dynamic scaling
Config Server	fetch properties	central config mgmt	change without redeploy
@RefreshScope	refresh bean	hot config reload	update limits

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10) Resilience (Circuit Breaker / Retry / Rate Limit)

Library/Class	Key Methods	Use Case	Example
Resilience4j CircuitBreaker	decorateSupplier(), executeSupplier()	avoid failures cascade	payment call protection
Resilience4j Retry	retry config	retry transient errors	network glitch
Bulkhead	limit concurrency	prevent overload	API overload protection
RateLimiter	limit traffic	stop abuse	OTP sending limit

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11) Messaging (Kafka / RabbitMQ)

Kafka (most common)

Class / Annotation	Key Methods	Real-world Use Case	Example
KafkaTemplate<K,V>	send(topic,msg), sendDefault()	publish events	UserCreatedEvent
@KafkaListener	consume messages	consume events	user provisioning

ConsumerRecord<K,V>	key(), value(), headers()	read metadata	tracing, offset
Acknowledgment	acknowledge()	manual commit	exactly once-like patterns
ProducerRecord	create record	custom headers	add correlation id

RabbitMQ (common)

Class	Methods	Use Case	Example
RabbitTemplate	convertAndSend()	send message	order created
@RabbitListener	consume	async processing	Billing

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12) Security (JWT / OAuth2 / Spring Security)

Class / Annotation	Methods	Use Case	Example
SecurityFilterChain	config DSL	define auth rules	protect endpoints
Authentication	getName(), getAuthorities()	logged-in user	roles
JwtDecoder	decode()	JWT verify	validate token
UserDetailsService	loadUserByUsernam e()	auth	load from DB
PasswordEncoder (BCryptPasswordEncod er)	encode(), matches()	secure password	store encrypted
@PreAuthorize	SpEL access	role-based access	@PreAuthorize ("hasRole('ADMIN')")
GrantedAuthority	getAuthority()	permissions	ROLE_USER

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13) Logging + Observability (Production MUST)

Class/Tool	Methods	Use Case	Example
SLF4J Logger	info(), debug(), warn(), error()	production logs	request tracing
MDC	put(), get(), clear()	correlation id logging	track 1 request across services
Micrometer MeterRegistry	counter(), timer()	metrics	latency monitoring
@Timed	annotation	measure API times	endpoint performance
@Observed	trace ops	distributed traces	service calls
OpenTelemetry	span/trace	tracing	Jaeger/Tempo

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14) Scheduling + Background Tasks

Class/Annotation	Key Methods	Use Case	Example
@Scheduled	cron/fixed rate	cleanup jobs	delete temp files
TaskScheduler	schedule tasks	dynamic schedules	schedule retry
ExecutorService	submit(), invokeAll()	parallel batch jobs	batch processing

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15) Testing (Used Daily)

Class	Methods	Use Case	Example
JUnit @Test	tests	unit tests	service test
Assertions	assertEquals(), assertThrows()	validations	exception check
Mockito	when(), thenReturn(), verify()	mocking	mock repository
@SpringBootTest	integration tests	full app test	microservice boot

@WebMvcTest	controller test	API test	endpoint behavior
MockMvc	perform(), andExpect()	test API response	200, JSON checks
Testcontainers	containers	real DB/Kafka test	Postgres test

✅ **Real Microservice Code Example (Short)**

Event-driven + REST + DB (Example use case: Order Service)

Controller

```
@PostMapping("/orders")
public ResponseEntity<OrderResponse> create(@Valid @RequestBody CreateOrderRequest req) {
   return ResponseEntity.status(201).body(orderService.createOrder(req));
}
```

Service

```
@Transactional
public OrderResponse createOrder(CreateOrderRequest req) {
   OrderEntity saved = orderRepo.save(OrderEntity.from(req));
   kafkaTemplate.send("order.created", new OrderCreatedEvent(saved.getId(), saved.getUserId()));
   return OrderResponse.from(saved);
}
```

Kafka Consumer in another microservice

```
@KafkaListener(topics = "order.created", groupId = "inventory-service")
public void onOrderCreated(OrderCreatedEvent event) {
   inventoryService.reserveStock(event.orderId());
}
```

If you want, I can also provide **the same table but version-wise**:

✅ Microservices in Java from **Spring Boot 1.x → 3.x**
✅ Spring Cloud versions + Netflix OSS → Resilience4j migration
✅ What classes/methods changed and what got deprecated

(in table format)